Augmentation and leakage

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

THE CONNECTED WEEK

Keep target validity,provenance and model evidence distinct.

Supply complete arrays and manifests;state the picture's conditional label rule and distinguish public audits from measured augmentation performance.

Keep one habit: Assess independent group/record/exclusion counts with checked keys and require reruns after a used split is repaired.

MAIN LESSON · SECTION 1

Prepare the connected mathematical investigation

> Keep explanation,shared practice and independent evidence distinct.

Target and main teaching plan

Primary target: Check whether a transformation preserves the task label

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min

> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Preparation and prerequisite

Printed lesson and workbook; paper and pencil

Calculator and optional local Python 3 editor; no packages or AI accounts needed

Use the fictional examples supplied; prepare the fresh retry before class

Using final outcomes to choose parameters or settings makes that set part of model selection rather than unseen **evidence** for the selected model.

Opening retrieval

Without opening the old lesson, explain one key idea from **Training** a small image model. Give an example and a mistake that the idea helps you avoid.

Look for: Trace a complete image model with fixed features and one trained weight

Reference example, not the only acceptable wording: Teacher explanation: d = 2, h1 = 2 and h2 = 0. Score is 0.9, **prediction** class 1 and squared error (0.9 - 1) squared = 0.01.

Without opening the old lesson, explain one key idea from Convolution as a local operation. Give an example and a mistake that the idea helps you avoid.

Look for: Compute a local weighted sum at each valid image position

Reference example, not the only acceptable wording: Teacher explanation: Window rows are [2, 0] and [1, 3]. Products are 2, 0, 0 and 3, giving **output** 5.

Without opening the old lesson, explain one key idea from Debugging shapes and gradients. Give an example and a mistake that the idea helps you avoid.

Look for: Check dimension contracts before computing a layer

Reference example, not the only acceptable wording: Teacher explanation: No. Each weight row must have the same length as x.

Confirm the intended feature schema, then supply a valid two-feature **input** or deliberately redesign and retrain the layer; do not silently discard a value.

Pictured end goal

By the end, I can explain **augmentation** and **leakage** using a traced example and a stated limitation.

▼ Earlier wording, alternatives and picture transcripts

Teach the learning cycle

Retrieval answer guidance

Training a small image model

Convolution as a local operation

Debugging shapes and gradients

For the compact alternative schedule, use two 45-minute sessions. This chapter explicitly revisits Week 23: different pixel arrays do not guarantee independent **augmentation** evaluation.

Keep its original toy-domain claim narrow and use a new grouped plan for the new experiment. Use fictional source IDs rather than student photos. No new **training** framework is needed.

Outcomes

Check whether a transformation preserves the task label

Keep related examples in one split using **source group**s

Compare **augmentation** fairly without inflating independent **evidence**

Preparation

Prerequisite answer

End goal for the pictorial case

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

Teach target validity and split validity as separate checks

> Keep the main binary task,P–T registry,pictured A–Dcounts and independent J–Oassignment distinct.

Main definitions,complete transform cases and guided key

Return to the binary image task: class 1 means the left column sum is strictly larger than the right column sum; equal sums belong to class 0. Let d be left sum minus right sum.

An **augmentation** is useful only under a stated task and transformation rule. More image files do not automatically mean more independent **evidence**.

Swapping rows preserves each column sum, so it preserves this task label for every valid image. Swapping columns changes d to -d. For nonzero d the class changes; for d = 0 it stays class 0.

Blindly copying the original label after every flip would create incorrect **training** targets.

A transformation may be valid for one task and invalid for another. A crop can remove the **evidence** needed for a label.

Choose allowed transformations using the task definition and **training evidence**, document the assumptions, and evaluate the choice on validation data.

This code assumes valid 2 by 2 binary lists. It constructs new lists instead of changing the **input**. The label function encodes the classroom task rule, not a general method for recovering unknown real-world labels.

If an allowed transformation changes a known target, the transformed example needs a justified new label. Do not invent labels for ambiguous cases.

Keep the source ID, transformation and resulting label with every derived image so a teacher can audit what happened.

Complete known-rule case	Original d / class	After row swap	After column swap
[[1, 0], [0, 0]]	1 /1	1 /1	–1 /0
[[0, 1], [0, 0]]	–1 /0	–1 /0	1 /1
Tie[[1, 0], [0, 1]]	0 /0	0 /0	0 /0; not complemented

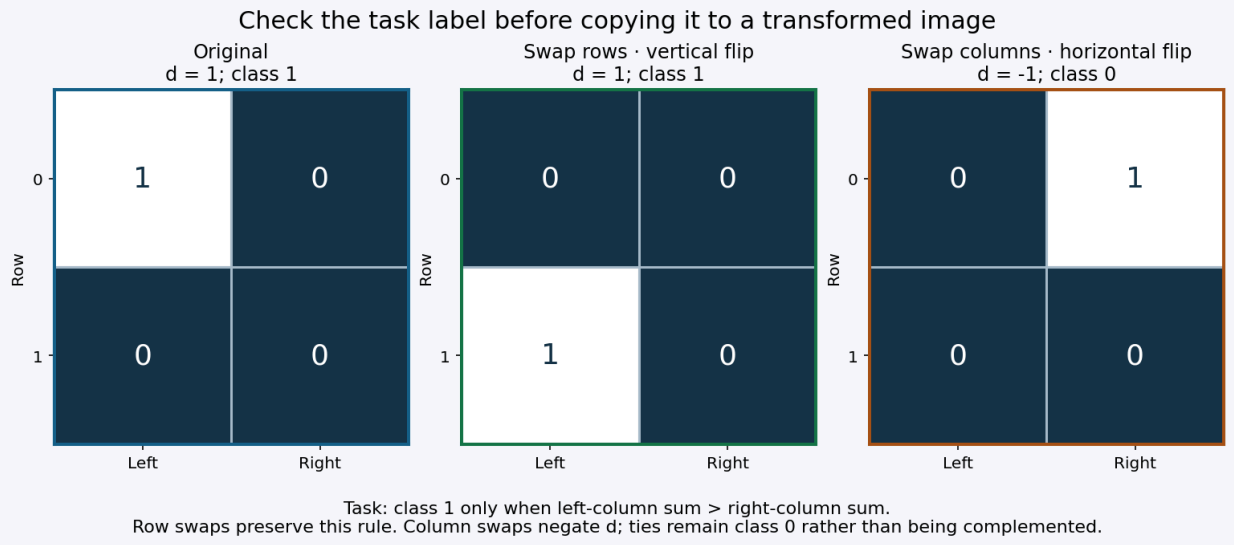
The label function is the supplied mathematical task, not a tool for inferring unknown real-world labels. Row swapping preserves each column sum. Column swapping negates d, so it changes the binary class only when d is nonzero. The original functions create new lists and leave their input unchanged.

The complete audit program later checks all 16 binary2×2 patterns for this rule and builds an explicitly fictional provenance registry. No augmented model is trained or improved score claimed by these audits.

Guided: the split is invalid for unseen-source evaluation because A2 derives from A regardless of filename. Keep the entire A group in **training** and replace the validation group, then rerun any affected selection.

Exit: budgets are 180 versus 60 updates, so they are not matched. Predeclare an equal update budget and record repetition policies; 36 records still represent only 12 original **source group**s.

Visible transformations and complete registry counts



Complete fictional original source · Binary image rows · Class under known rule · Assigned role · Retention policy

> **Complete fictional original source:** P
>
> **Binary image rows:** [1, 0]; [0, 0]

Class under known rule: 1

Assigned role: Training

Retention policy: Original and row swap

Complete fictional original source: Q

Binary image rows: [0, 1]; [0, 0]

Class under known rule: 0

Assigned role: Training

Retention policy: Original and row swap

Complete fictional original source: R

Binary image rows: [1, 0]; [1, 0]

Class under known rule: 1

Assigned role: Validation

Retention policy: Original only

Complete fictional original source: S

Binary image rows: [0, 1]; [0, 1]

Class under known rule: 0

Assigned role: Final

Retention policy: Original only

Complete fictional original source: T

Binary image rows: [1, 1]; [0, 0]

Class under known rule: 0

Assigned role: Final

Retention policy: Original only

Registry result	Source groups	Retained records	Derived evaluation records excluded
Training P/Q	2	4	0
Validation R	1	1	1
Final S/T	2	2	2
Total	5	7	3

Separate matched-update budget options

Separate budget comparison; batch size 1 · Records per epoch · Epochs · Updates · Source identities

> **Separate budget comparison; batch size 1:** 12 originals, eachwith 2 variants
> **Records per epoch:** 36
> **Epochs:** 5
> **Updates:** 180
> **Source identities:** 12

> **Separate budget comparison; batch size 1:** 12 originals only
> **Records per epoch:** 12
> **Epochs:** 5
> **Updates:** 60
> **Source identities:** 12

> **Separate budget comparison; batch size 1:** One declared equal-update baseline option
> **Records per epoch:** 12
> **Epochs:** 15
> **Updates:** 180

> **Source identities:** 12

> **Separate budget comparison; batch size 1:** Independent source count meaning
>
> **Records per epoch:** Variants are related to their originals
>
> **Epochs:** Record exposure policy
>
> **Updates:** Matched updates do not prove benefit
>
> **Source identities:** 36 files are not 36 independent subjects

Pictured local source groups and task-specific label caveat

Keep every augmented image with its original **source group**.

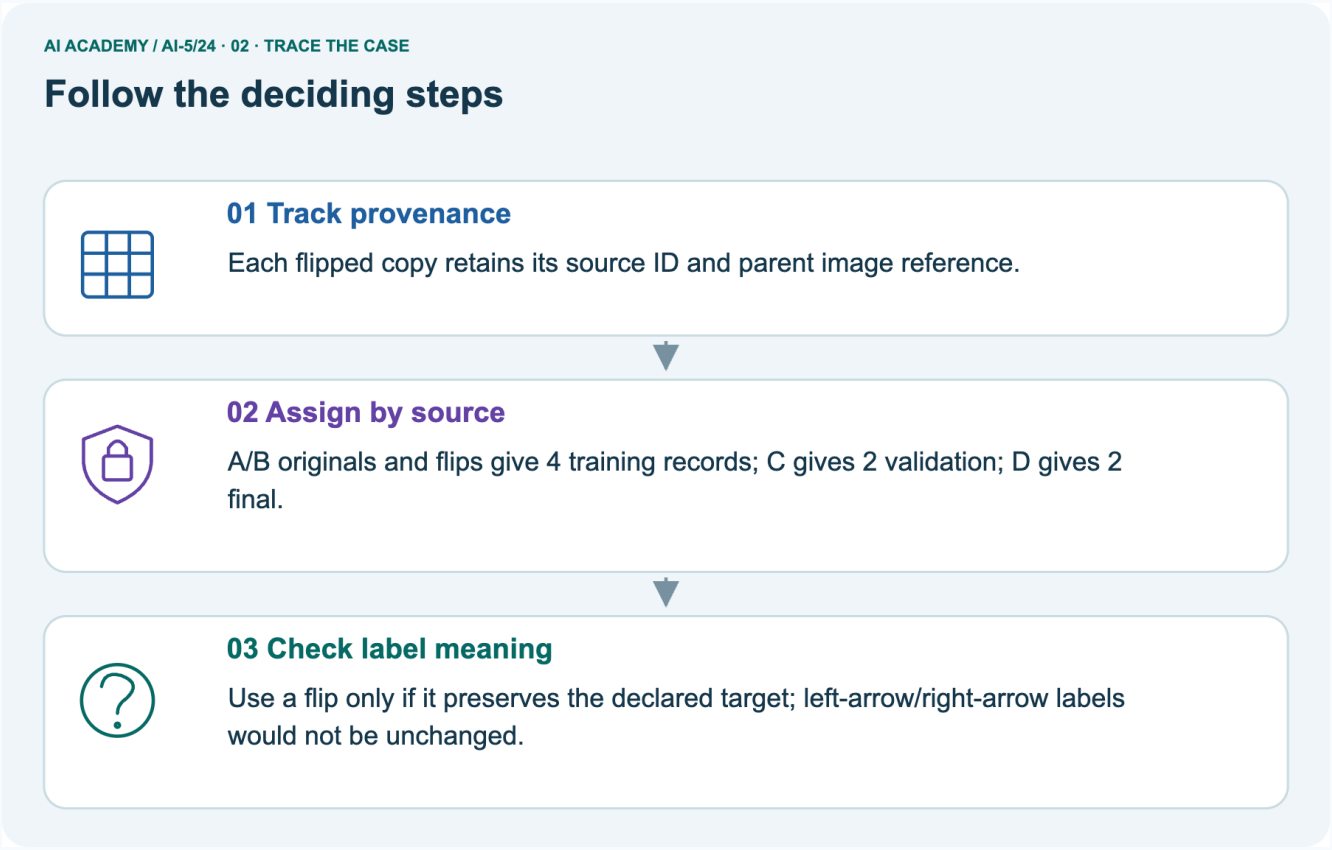
Fictional sources A, B, C, D each have original+horizontal-flip copies. A/B train, C validation, D final test.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

The picture is a separate provenance/counting example with local source names A–D, not the six image IDs from Week23. Each source has an original plus a horizontal-flip record. A/Btraining gives 4 records from 2 groups; Cvalidation and Dfinal each have 2 records from 1 group.

These counts do not certify label validity:for our left-versus-right target, a horizontal flip generally changes the class and needs its justified target. Moving D's derived record to training crosses the final source boundary. The new P–T registry below uses verified row swaps and retains only original evaluation records.

Trace cross-split lineage despite a new filename



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Track provenance	Each flipped copy retains its source ID and parent image reference.
2. Assign by source	A/B originals and flips give 4 training records; C gives 2 validation; D gives 2 final.
3. Check label meaning	Use a flip only if it preserves the declared target; left-arrow/right-arrow labels would not be unchanged.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

An augmented image remains related to its source. A horizontal flip changes the view, not the identity of the original example. Assign **source group**s to **training**, validation and final evaluation before producing copies.

Also check that the transformation preserves the target: flipping an arrow can change its correct direction label.

Augmentation validity is task-specific, and it cannot manufacture new independent plants or repair inadequate sampling.

Change: Move D's flipped copy into **training**.

Prediction: Final-test group separation fails.

Reason: The transformed D image is still derived from the held-out source, despite a new filename.

▼ Earlier wording, alternatives and picture transcripts

Term	Meaning
Augmentation	Creating transformed training examples under a declared procedure
Label preservation	A transformation leaving the correct target unchanged for the stated task
Source group	An identifier joining records that must stay together when splitting
Leakage	Information crossing an evaluation boundary in a way that undermines the intended independent check

Every variant stays with its source group

SOURCE GROUP	ASSIGNED SPLIT	DERIVED RECORDS
A	Training	A original and A row swap
B	Validation	B original only
C	Final test	C original only

Assign source groups before generating training variants. A new filename does not make a derived image independent. Evaluation examples follow a fixed declared protocol, not random training augmentation.

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Every variant stays with its source group

SOURCE GROUP	ASSIGNED SPLIT	DERIVED RECORDS
A	Training	A original and A row swap
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C	Final test	C original only

Assign source groups before generating training variants. A new filename does not make a derived image independent. Evaluation examples follow a fixed declared protocol, not random training augmentation.

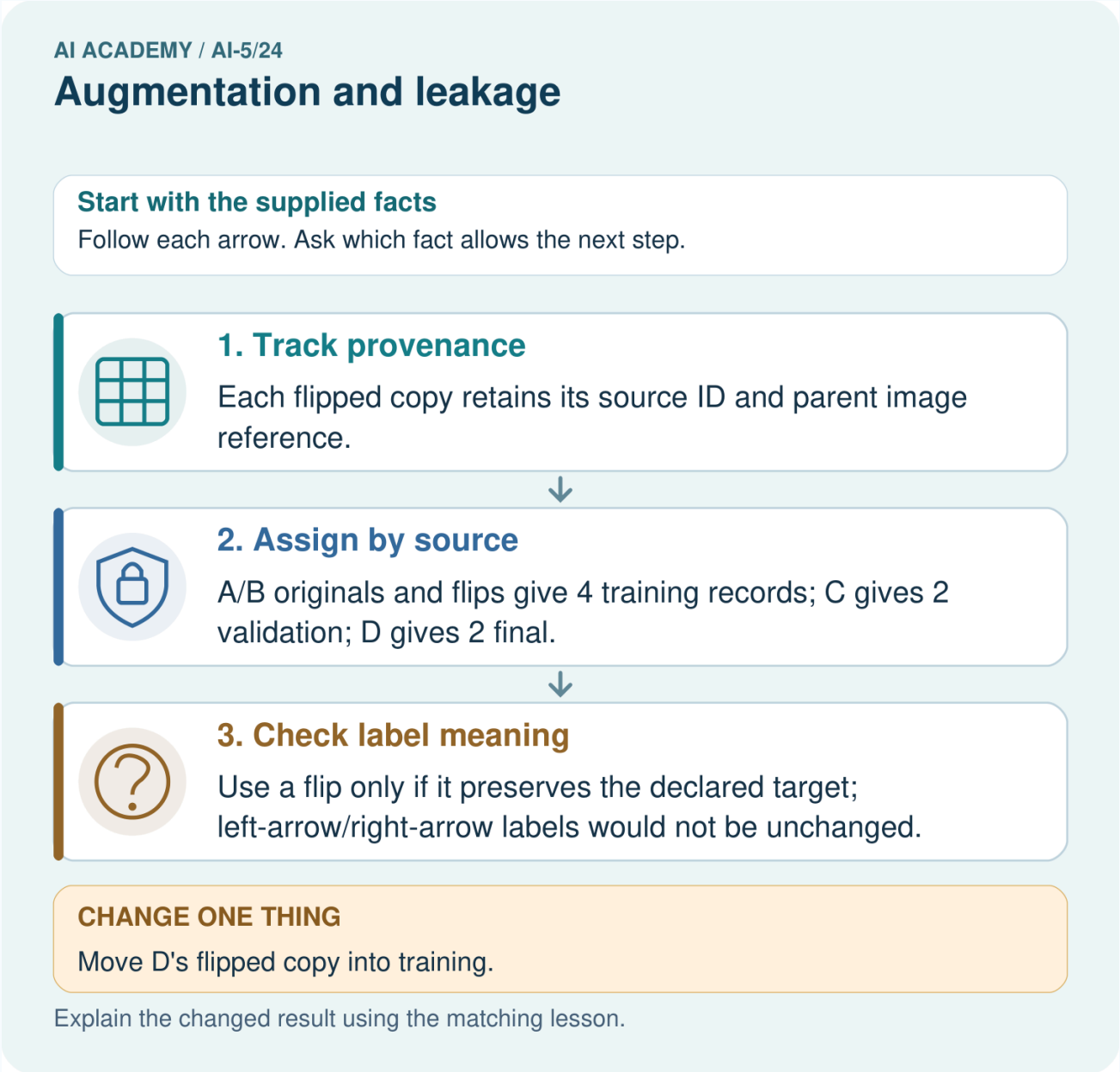
A training image has source ID A. Its row-swapped copy is called A2 and placed in validation. No filenames match across splits. Is the split valid for evaluating unseen source images? Describe the repair.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Augmentation and leakage

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

MAIN LESSON · SECTION 3

Model a step,then remove help

> Collect each learner’s reasoning before showing a key.

Current completion task and checked reasoning

Completion task: For rows [1, 0], [0, 1], compute d and class before and after a column swap. Why is always replacing a label by 1 minus the label incorrect?

Teacher check: Teacher explanation: Both differences are 0 and both labels are class 0. Complementing class 0 would give class 1, which is wrong for ties.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for **Augmentation** and **leakage**. Proposed explanation: “Every image flip preserves the label”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: Validity depends on the task; column swaps change non-tie labels here.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictured reasoning and changed case

How many independent **source group**s are in the final test? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

One, D, with two related records. Count both **source group**s and image rows; never describe its flip as a new independent source. Changed-case check: Final-test group separation fails.

The transformed D image is still derived from the held-out source, despite a new filename.

Changed-case reasoning: Final-test group separation fails. The transformed D image is still derived from the held-out source, despite a new filename.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Accessible support and extension

Use See how it works and Follow a worked example. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Use cards with source ID and variant ID in separate columns. Provide a split ledger and distinguish original count, variant count and total record count.

Extension: enumerate all sixteen binary images to verify row-swap preservation, count column-swap label changes, and identify the equality cases that defeat automatic label complementation.

▼ Earlier wording, alternatives and picture transcripts

Time or stage	Teacher action
Session 1 start to 7 minutes	Retrieve frozen evaluation and restate the strict left-column task label.
7 to 18 minutes	Trace row and column swaps, then complete W1 and W2 including the equality boundary.
18 to 32 minutes	Run or hand-trace the transformation code and inspect source preservation.
32 to 45 minutes	Use the visual and guided case to keep lineage groups together before augmentation.
Session 2 start to 8 minutes	Audit the Week 23 transformation relationships and complete W3.
8 to 20 minutes	Distinguish source counts from record counts and work through the update-budget example.
20 to 35 minutes	Collect W4 independently, requiring source counts, retained counts and exclusions.
35 to 45 minutes	Use W5 and the exit response to propose a fair comparison, then administer the fresh retry.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the complete supplied investigation

> Separate predictions,actual runs and paper traces.

Complete inputs,expected results and limits

W1 row/column rules and tie exception as supplied. W2 tie d0/class 0 before/after, not complement 1. W3P crop crosses Ptraininggroup; repair provenance/splits and rerun affected fitting/selection, not rename while preserving a clean-score claim. W5 budgets96 versus32; match a predeclared update budget/exposure policy. P–T registry has 7 retained/3 excluded and code audits recordedgroups/exactpixels separately.

Independent J/K/Ltrain3 groups×4 records=12; Mvalidation1 group/1 original; N/Ofinal2 groups/2 originals; total 15 retained, 9 derived evaluation rows excluded. Small group counts still bound evidence.

Diagnose the current misconceptions

Every image flip preserves the label: Validity depends on the task; column swaps change non-tie labels here.

Different filenames prove independence: Lineage, shared subjects and identical content can connect different filenames.

More augmented files mean more independent examples: Variants retain dependence on their source; report both group and record counts.

Use Run the investigation for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Choose accessible support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.

What the response shows	Next teaching action
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ Earlier wording, alternatives and picture transcripts

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the new schema or calculation transfer

> Keep the selected answer private until after a first attempt.

Selected independent task and checked answer

Selected case: Six fictional **source group**s J, K, L, M, N and O each have one original and three derived records. Assign J, K and L to **training**, M to validation and N, O to final test.

Keep all four records per **training** group but only originals for evaluation. Give source and retained-record counts per split, total retained records and excluded derived records.

Teacher explanation: **Training**: 3 groups and 12 records. Validation: 1 group and 1 record. Final test: 2 groups and 2 records. Total retained records are 15. Nine derived evaluation records are excluded.

No group crosses splits; the small group counts still limit evaluation **evidence**.

Judge explanation,trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one usable repair

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Check the label after a transformation

For rows [0, 1], [0, 1], compute d and class before and after swapping rows and after swapping columns.

Teacher explanation: Original d = -2 and class 0. Row swapping leaves d = -2 and class 0. Column swapping gives rows [1, 0], [1, 0], d = 2 and class 1.

W2 Keep the equality boundary

For rows [1, 0], [0, 1], compute d and class before and after a column swap. Why is always replacing a label by 1 minus the label incorrect?

Teacher explanation: Both differences are 0 and both labels are class 0. Complementing class 0 would give class 1, which is wrong for ties.

W3 Audit related records

Training contains P original and Q original. Validation contains P crop and R original. Final test contains S original. P crop is derived from P original.

Identify the overlap and explain what must be rerun if this validation set already selected a model.

Teacher explanation: Source P crosses **training** and validation. Keep all P records in one split and choose distinct evaluation groups.

Rerun **training** and model selection on the corrected plan; the old validation score does not describe the repaired experiment.

W5 Match a training budget

Eight source images each have an original plus two variants. Compare four epochs of these 24 records with four epochs of the eight originals, using one-example batches.

Give both update counts and one way to match budgets without using final outcomes.

Teacher explanation: Augmented run uses 96 updates; original-only run uses 32. Predeclare 96 updates for both, equivalent to 12 passes over the originals for the baseline.

Record repetition and **augmentation** policies, use validation for choices and keep final evaluation reserved.

▼ Earlier wording, alternatives and picture transcripts

Guided and exit questions

Guided: the split is invalid for unseen-source evaluation because A2 derives from A regardless of filename. Keep the entire A group in **training** and replace the validation group, then rerun any affected selection.

Exit: budgets are 180 versus 60 updates, so they are not matched. Predeclare an equal update budget and record repetition policies; 36 records still represent only 12 original **source group**s.

W1 Check the label after a transformation

W2 Keep the equality boundary

W3 Audit related records

W4 Design a grouped split independently

Six fictional **source group**s J, K, L, M, N and O each have one original and three derived records. Assign J, K and L to **training**, M to validation and N, O to final test.

Keep all four records per **training** group but only originals for evaluation. Give source and retained-record counts per split, total retained records and excluded derived records.

Teacher explanation: **Training**: 3 groups and 12 records. Validation: 1 group and 1 record. Final test: 2 groups and 2 records. Total retained records are 15. Nine derived evaluation records are excluded.

No group crosses splits; the small group counts still limit evaluation **evidence**.

W5 Match a training budget

Independent assessment and fresh retry

W4 requires source counts 3, 1, 2 and retained counts 12, 1, 2, with 15 total records and 9 excluded derived records. Fresh retry: five groups A through E each have an original plus two variants.

Train on A and B using all records, validate on C original and test on D and E originals. Source counts are 2, 1, 2; retained records are 6, 1, 2; total 9, with 6 derived evaluation records excluded.

Three augmented epochs at batch size 1 use 18 updates, compared with 6 for three epochs over two originals.

Assess independent transfer

Selected case answer

Feedback that the learner uses

Connect,support and revisit the mathematics

> Use the saved evidence to choose the next step.

Project and next-week bridge

Add a labelled **augmentation** and **leakage evidence** sheet to Image-pattern Network Investigation. Show how you can check whether a transformation preserves the task label.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Use cards with source ID and variant ID in separate columns. Provide a split ledger and distinguish original count, variant count and total record count.

Extension: enumerate all sixteen binary images to verify row-swap preservation, count column-swap label changes, and identify the equality cases that defeat automatic label complementation.

Week 25 introduces sequence representations and the information preserved by order.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Delayed return and support record

Suggested return points in this level: ai-5/25, ai-5/27, ai-5/30. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

▼ Earlier wording, alternatives and picture transcripts

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 7

Reference · Alternative planning checklist

> Use alongside the main route when helpful.

Planning,assessment and project reminders

Teaching target: Check whether a transformation preserves the task label

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
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> **Session:** Session 3 · Investigate and transfer · 45 min
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> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

Before class, solve the supported practice: For rows [1, 0], [0, 1], compute d and class before and after a column swap. Why is always replacing a label by 1 minus the label incorrect?

Reasoned practice answer: Teacher explanation: Both differences are 0 and both labels are class 0. Complementing class 0 would give class 1, which is wrong for ties.

Check every learner before discussion: Use the worked case for **Augmentation** and **leakage**. Proposed explanation: “Every image flip preserves the label”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: Validity depends on the task; column swaps change non-tie labels here.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Move D’s flipped copy into **training**. Require a **prediction** and a reason before comparing. Expected result: Final-test group separation fails.

Reason: The transformed D image is still derived from the held-out source, despite a new filename.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback,labs and repair paths**

REFERENCE · SECTION 8

Reference · Feedback,labs and repair paths

> Retain the first attempt and the evidence used for a repair.

Feedback and record guidance

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

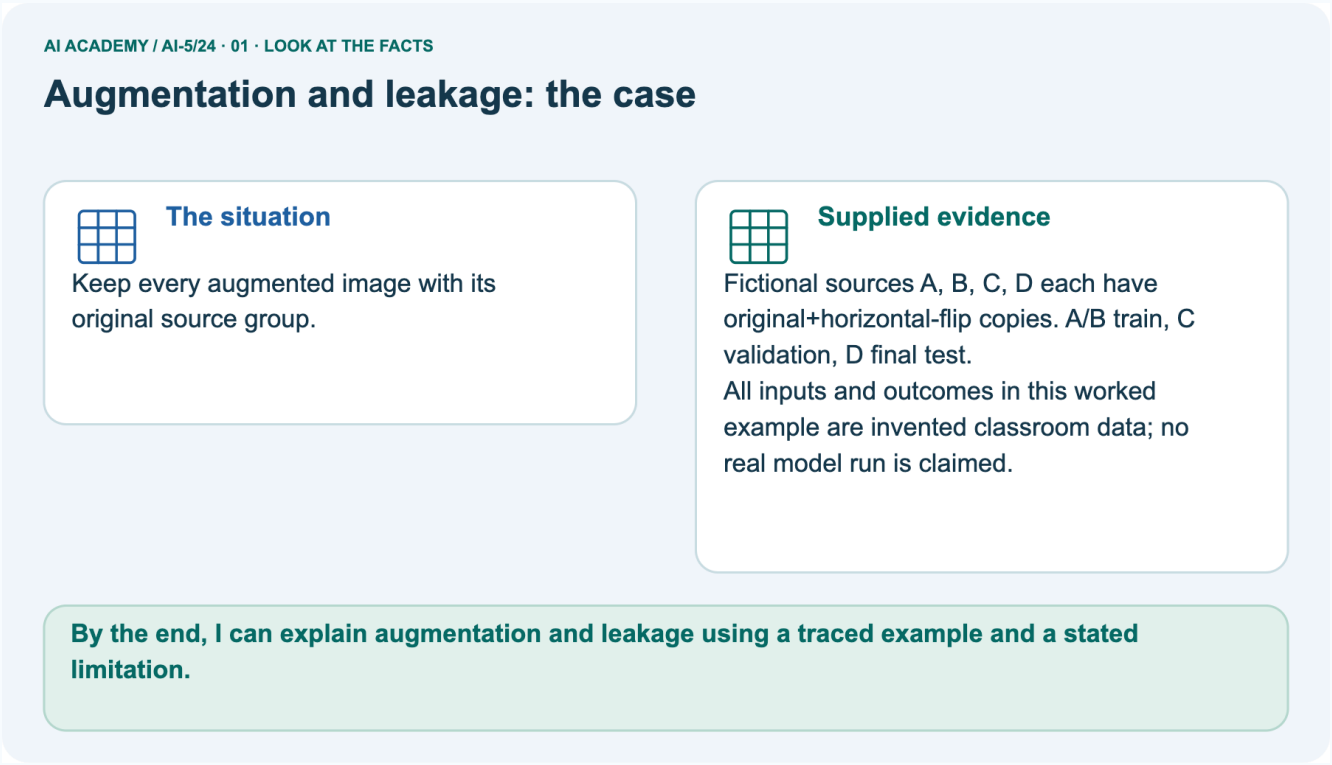
▼ **Reference · Teach with the six pictures**

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Choose one picture for a particular reasoning move.

01. Augmentation and leakage: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

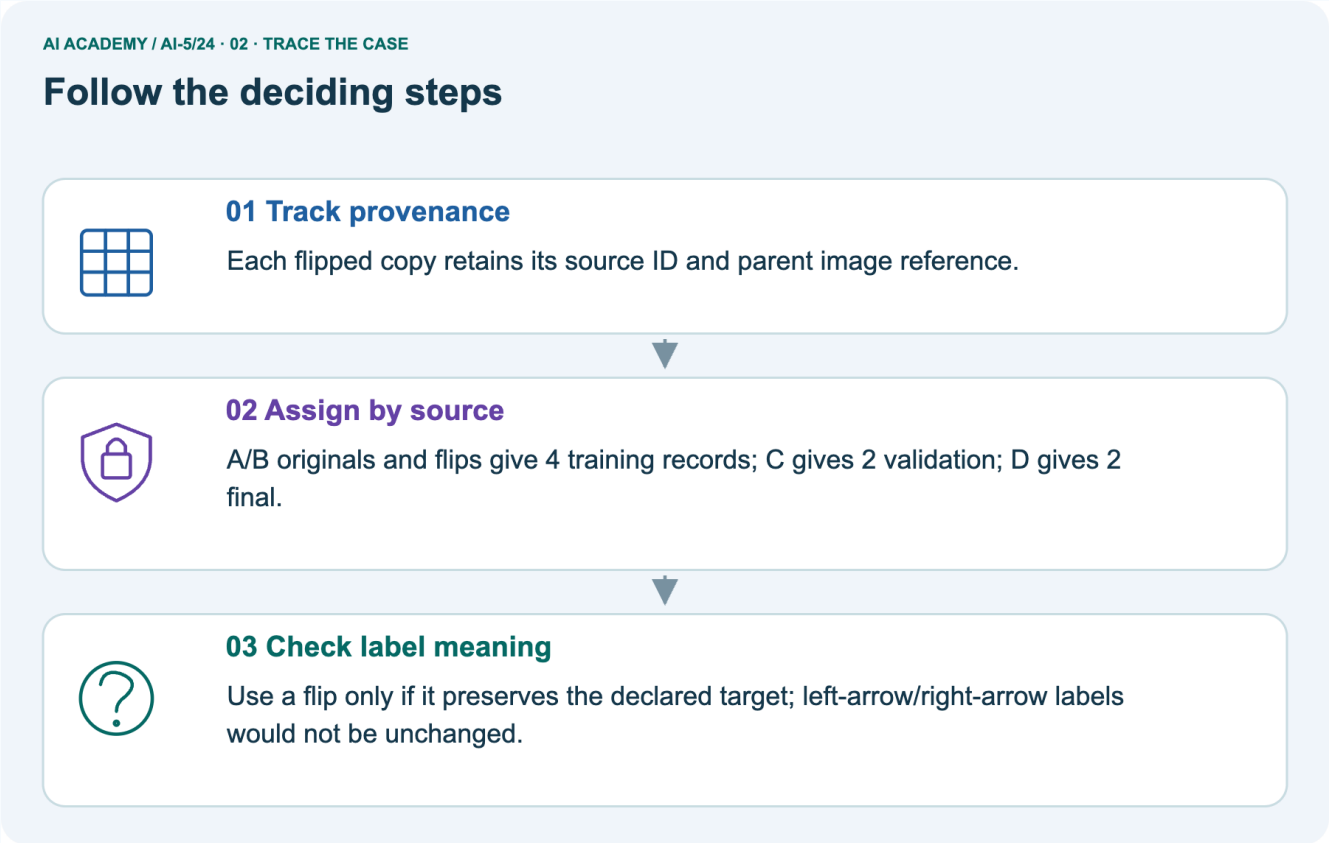
Keep every augmented image with its original **source group**. Supplied facts: Fictional sources A, B, C, D each have original+horizontal-flip copies. A/B train, C validation, D final test.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



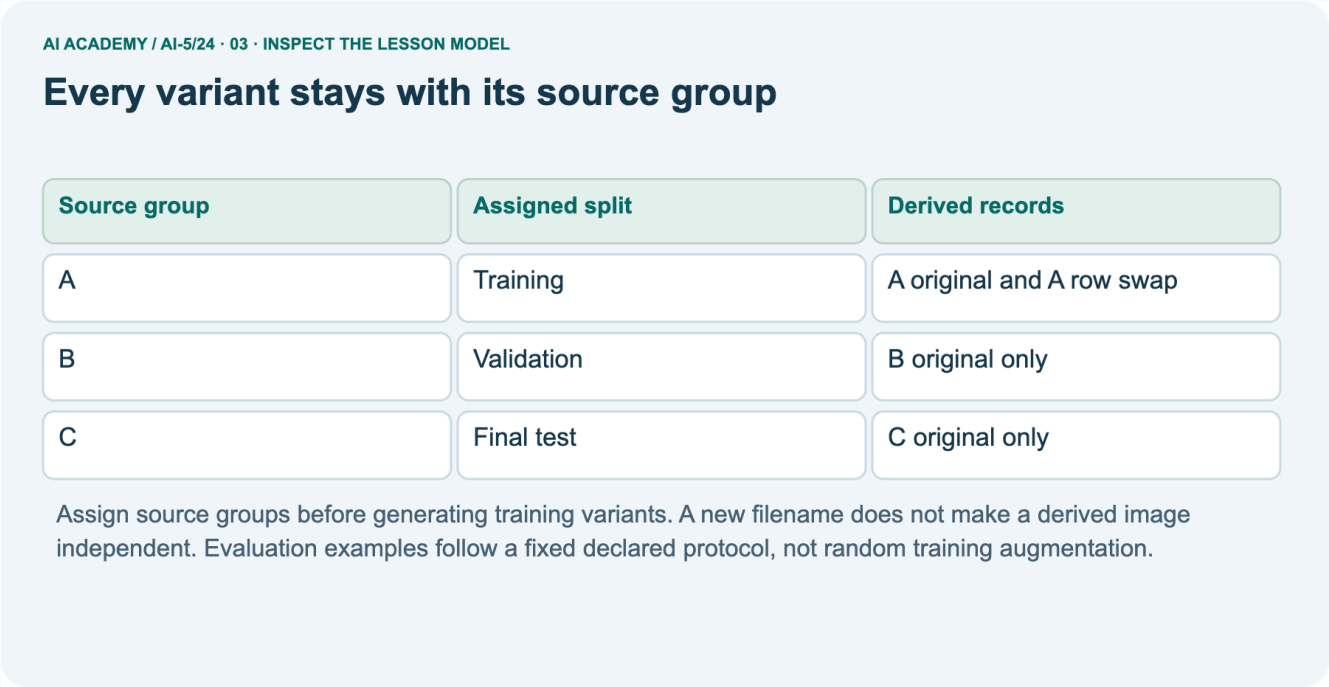
Follow the labelled picture with the matching explanation. Open the vector diagram

Track provenance: Each flipped copy retains its source ID and parent image reference. → Assign by source: A/B originals and flips give 4 **training** records; C gives 2 validation; D gives 2 final. → Check label meaning: Use a flip only if it preserves the declared target; left-arrow/right-arrow labels would not be unchanged.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Every variant stays with its source group



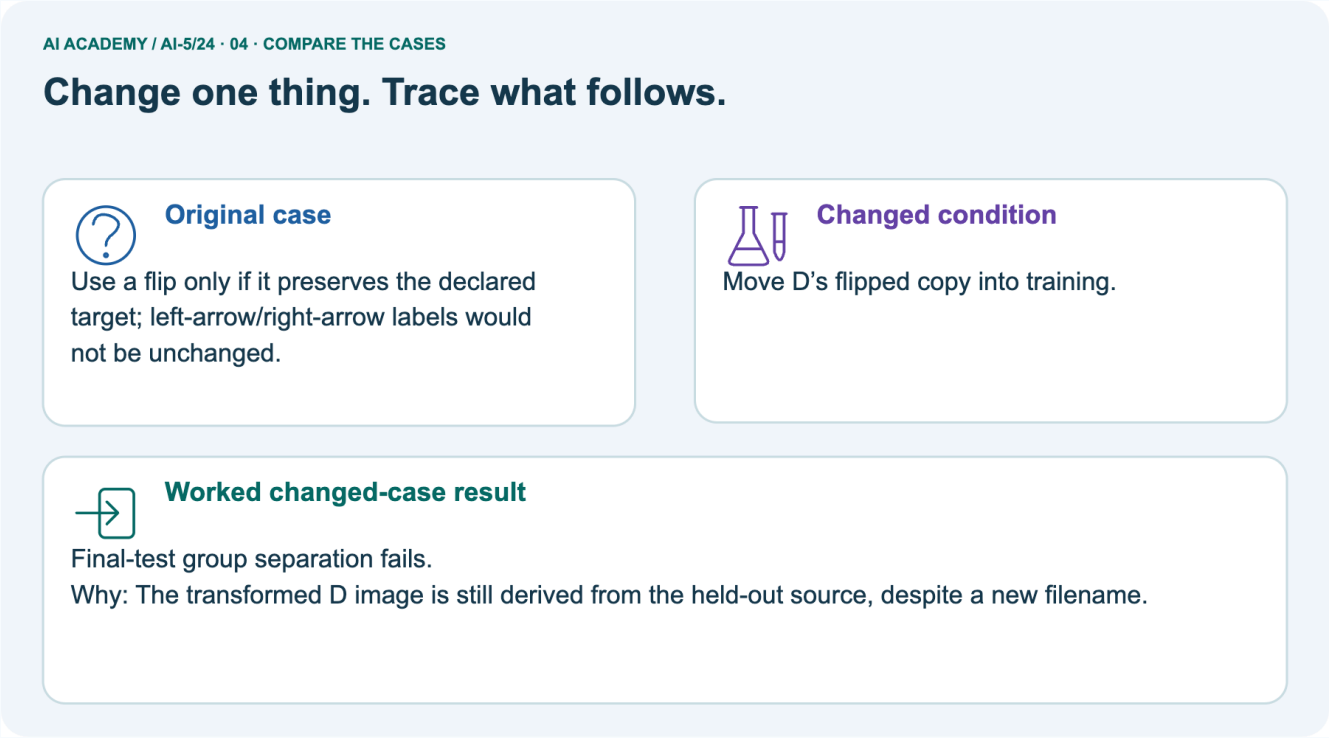
Follow the labelled picture with the matching explanation. Open the vector diagram

Every variant stays with its **source group**. A | **Training** | A original and A row swap. B | Validation | B original only. C | Final test | C original only

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Use a flip only if it preserves the declared target; left-arrow/right-arrow labels would not be unchanged. Changed condition: Move D's flipped copy into **training**.

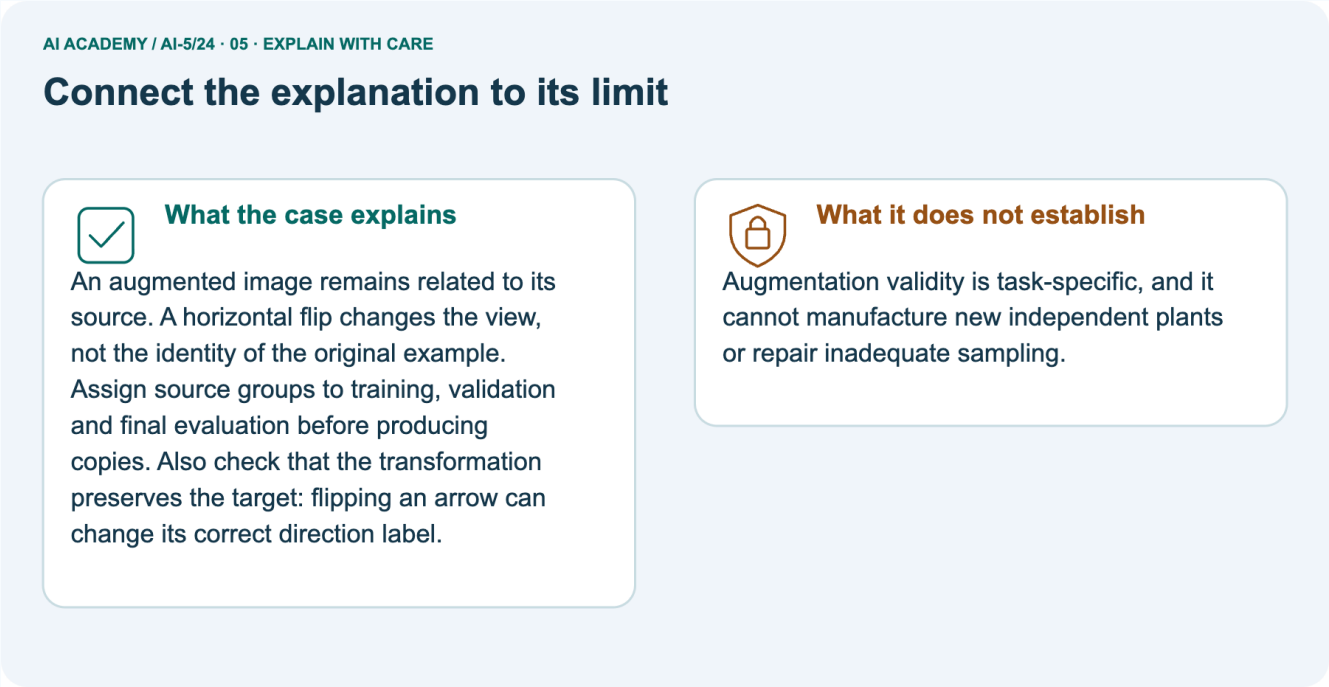
Expected result: Final-test group separation fails. Reason: The transformed D image is still derived from the held-out source, despite a new filename.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: The transformed D image is still derived from the held-out source, despite a new filename.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

An augmented image remains related to its source. A horizontal flip changes the view, not the identity of the original example. Assign **source group**s to **training**, validation and final evaluation before producing copies.

Also check that the transformation preserves the target: flipping an arrow can change its correct direction label.

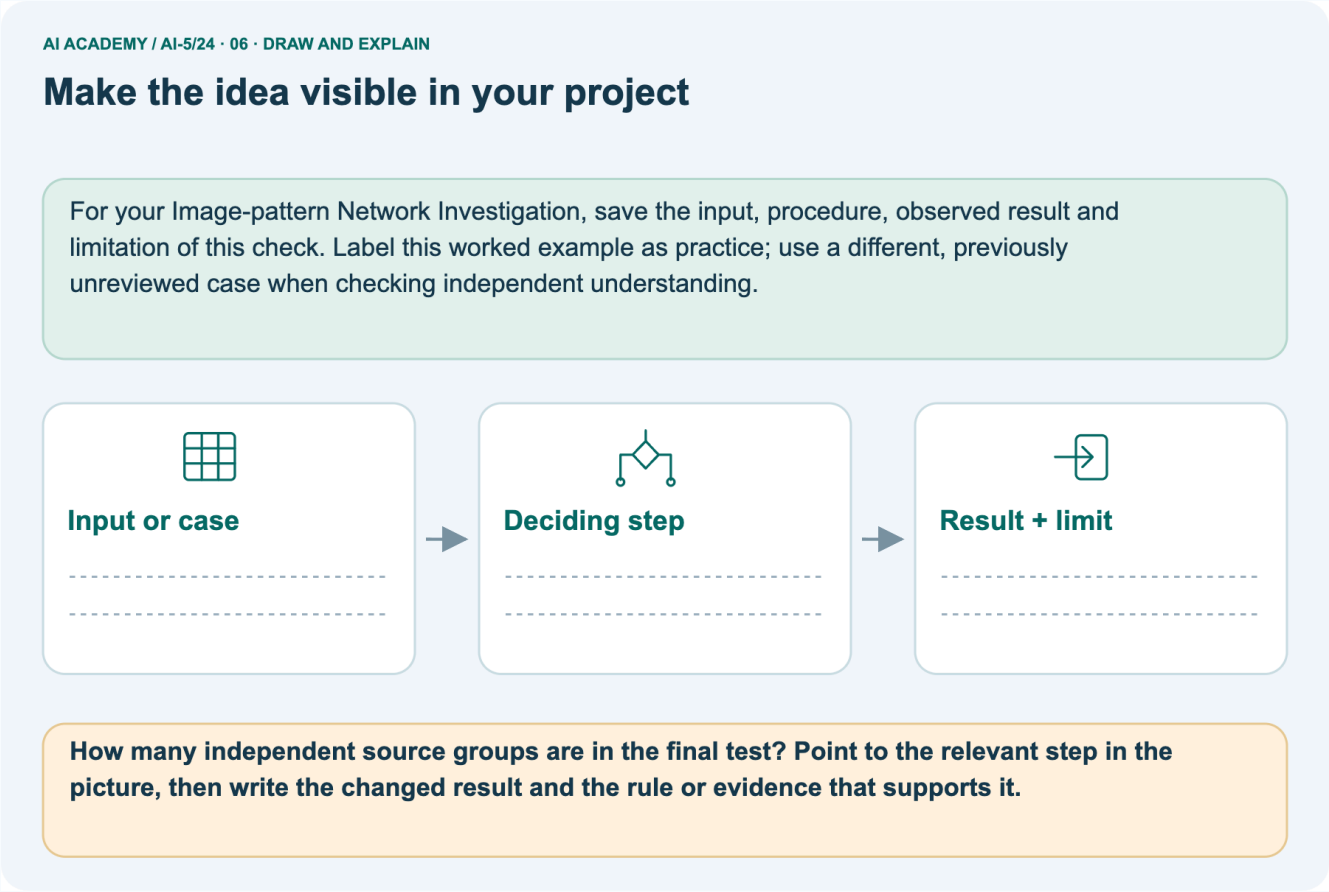
Limit: **Augmentation** validity is task-specific, and it cannot manufacture new independent plants or repair inadequate sampling.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: **Augmentation** validity is task-specific, and it cannot manufacture new independent plants or repair inadequate sampling.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: How many independent **source group**s are in the final test? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: One, D, with two related records. Count both **source group**s and image rows; never describe its flip as a new independent source. Changed-case check: Final-test group separation fails.

The transformed D image is still derived from the held-out source, despite a new filename.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Augmentation and leakage: the case

02. Follow the deciding steps

03. Every variant stays with its source group

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.